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The purpose of this document is to present the setup and configuration of the Splunk infrastructure required to build a SOC lab environment. The lab will provide a centralised platform for collecting and analysing security data, monitoring system activity, detecting potential threats, and investigating security events.

Splunk Installation

Splunk Server will be hosted on the Ubuntu VM that was previously configured to be in the same local network as other machines later used to forward data to Splunk.

```
lab_admin@SplunkServer:~$ ls -ld /opt/splunk
drwxr-xr-x 10 splunk splunk 4096 Aug 25 12:09 /opt/splunk
lab_admin@SplunkServer:~$ id splunk
uid=1001(splunk) gid=1001(splunk) groups=1001(splunk)
lab_admin@SplunkServer:~$
```

After downloading the installation package, it successfully installed as shown above.

After accepting the licence and creating an account, Splunk is ready to be accessed.

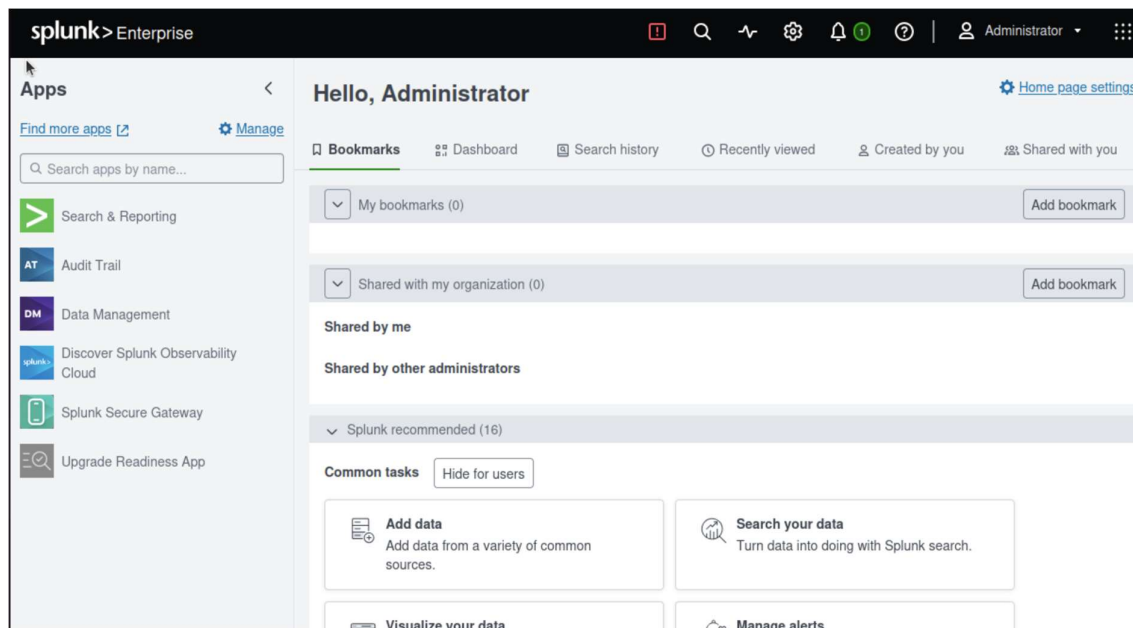
The Splunk web interface is at <http://SplunkServer:8000>

```
lab_admin@SplunkServer:~$
```

 splunkserver:8000/en-US/account/login?return_to=%2Fen-US%2F 

 > enterprise

Username	Password	Sign In
---------------------------------------	---------------------------------------	--

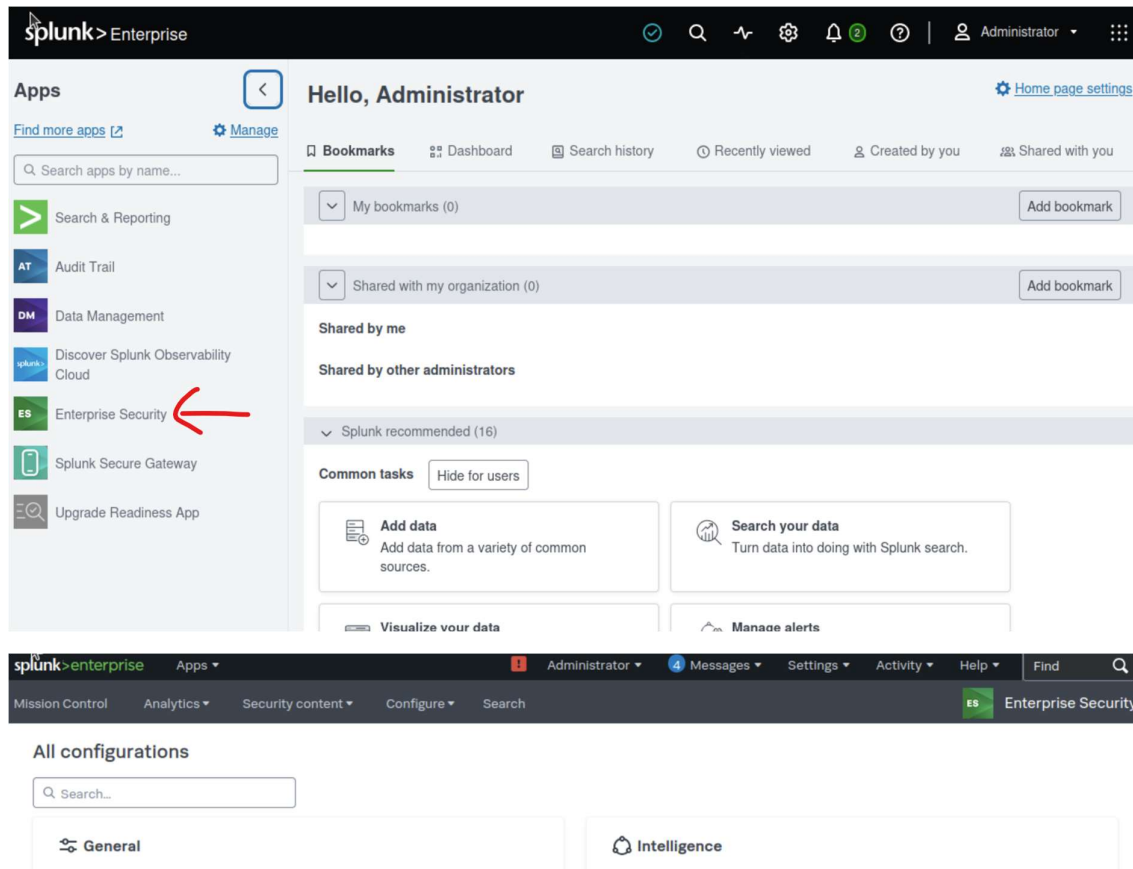


Splunk Enterprise Security (ES)

My next step was to install Splunk Enterprise Security (ES), which is an add-on to Splunk Enterprise that provides advanced security monitoring, threat detection, investigation, and incident response capabilities.

```
lab_admin@SplunkServer:/$ sudo -u splunk /opt/splunk/bin/splunk install app /opt
/splunk/splunk-enterprise-security_851.spl
WARNING: Server Certificate Hostname Validation is disabled. Please see server.c
onf/[sslConfig]/cliVerifyServerName for details.
App '/opt/splunk/splunk-enterprise-security_851.spl' installed
lab_admin@SplunkServer:/$ sudo -u splunk /opt/splunk/bin/splunk restart
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
.....
Stopping splunk helpers...
Done.

Splunk> Winning the War on Error
```



Splunk Configuration

In this part, I adjusted Splunk configuration:

- Enabled to run at boot time

```
lab_admin@SplunkServer:/$ sudo /opt/splunk/bin/splunk enable boot-start -user splunk
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
lab_admin@SplunkServer:/$
```

- Configured receiving data:

splunk>Enterprise [1] [Q] [↶] [⚙] [🔔 4] [?] | [👤] Administrator ▾

Forwarding and receiving

Forward data
Set up forwarding between two or more Splunk instances.

Type	Actions
Forwarding defaults	
Configure forwarding	+ Add new

Receive data
Configure this instance to receive data forwarded from other instances.

Type	Actions
Configure receiving	+ Add new

Add new

Forwarding and receiving » Receive data » Add new

Configure receiving
Set up this Splunk instance to receive data from forwarder(s).

Listen on this port *
For example, 9997 will receive data on TCP port 9997.

[Cancel](#) [Save](#)

Receive data
Forwarding and receiving » Receive data [New Receiving Port](#)

Successfully saved "9997".

Showing 1-1 of 1 item

filter [Q] 25 per page ▾

Listen on this port ▾	Status ▾	Actions
9997 ✓	Enabled Disable	Delete

*** Note

Joanna, 04/09/2026

To improve the performance of the Splunk Server, I changed Ubuntu from the graphical boot target to CLI-only mode. This reduced unnecessary resource usage while keeping Splunk running automatically and accessible through its web interface. After rebooting, I confirmed that Splunk started successfully and the web interface was noticeably more responsive.

Splunk Forwarders

In this part, I will install Splunk Universal Forwarders on the host VMs within my local lab network. The forwarders will collect log data from each host and send it to the Splunk server, where the data can be indexed, searched, monitored, and investigated.

Windows Server

On Windows Server 2022, I downloaded Splunk Forwarder from Splunk official website.

https://www.splunk.com/en_us/download/universal-forwarder.html

Splunk Universal Forwarder 10.4.2

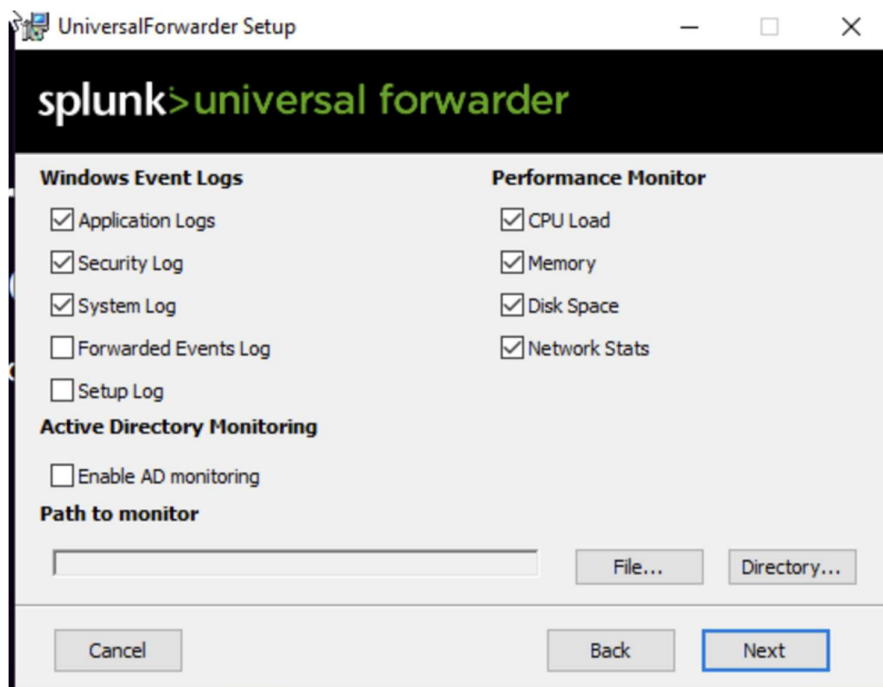
Universal Forwarders provide reliable, secure data collection from remote sources and forward that data into Splunk software for indexing and consolidation. They can scale to tens of thousands of remote systems, collecting terabytes of data.

Choose Your Installation Package

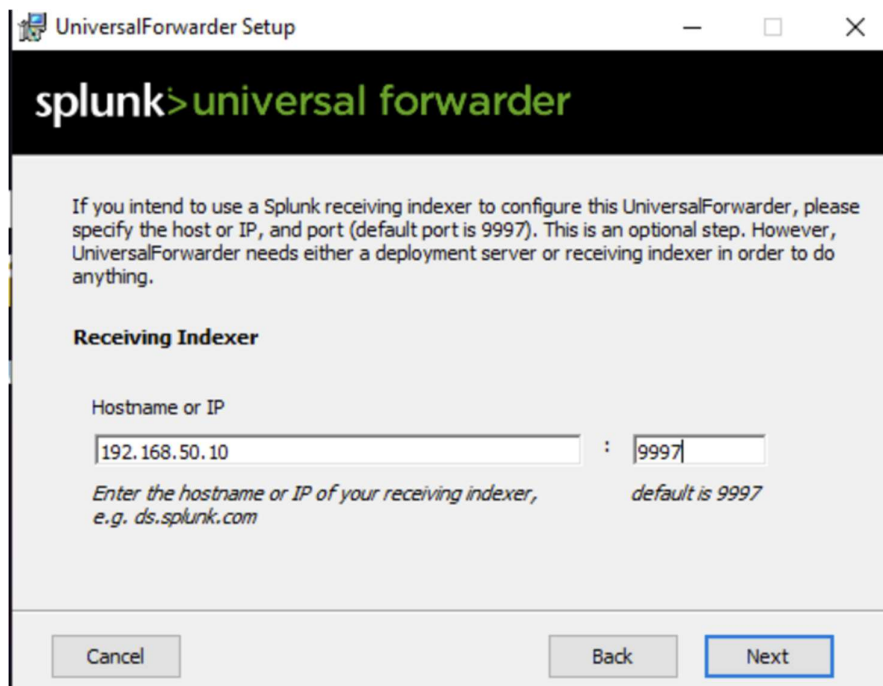
Platform	Architecture	Operating System	File Format	Size	Download Now	Copy wget link	More
Windows	64-bit	Windows 10, 11 Windows Server 2019, 2022, 2025	.msi	151.31 MB	Download Now	Copy wget link	More
	32-bit	Windows 10	.msi	64.94 MB	Download Now	Copy wget link	More

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) | [All Other Downloads](#)

During installation of the forwarder, I have chosen what data type I want to be forwarded to Splunk server:



I also specified Splunk Server's Receiving Indexer IP address and the port that splunk is listening on:





To verify that the forwarder is successfully sending data to Splunk, I run an SPL query to see the statistics on host and sourcetypes. As seen below, the installation was successful.

The screenshot shows the Splunk Enterprise web interface. The browser address bar shows 'splunkserver:8000/en-US/app/search/search?q=search index%3D* host%3D* earliest%3D-15m%0'. The search bar contains the query: `index=* host=* earliest=-15m | stats count by host sourcetype`. Below the search bar, it shows '1,513 events (8/25/26 8:34:31.000 PM to 1/19/38 3:14:07.000 AM)'. The results are displayed in a table with columns 'host', 'sourcetype', and 'count'.

host	sourcetype	count
SplunkServer	stash	1
Win-Serv-22	Perfmon:Available Memory	20
Win-Serv-22	Perfmon:CPU Load	40
Win-Serv-22	Perfmon:Network Interface	73
Win-Serv-22	WinEventLog:Application	103
Win-Serv-22	WinEventLog:Security	1008
Win-Serv-22	WinEventLog:System	268

My next step is to install Splunk Add-on for Microsoft Windows to correctly interpret and normalize Windows logs, allowing them to be effectively analysed and used by Splunk Enterprise Security.



Splunk Add-on for Microsoft Windows

[Install](#)

*** Important: Read upgrade instructions and test add-on update before deploying to production *** The Splunk Add-on for Windows 5.0.0 introduced breaking changes. If you are upgrading from a version of the Splunk Add-on for Windows that is earlier than 5.0.0, you must follow the documented upgrade instructions to avoid data loss. A best practice is to test the upgraded version...[More](#)

Category: [IT Operations](#), [Security](#), [Fraud & Compliance](#) | Author: [Splunk LLC](#) | Downloads: 771542 | Released: 0 months ago | Last Updated: 0 months ago | [View on Splunkbase](#)

New Search SPL ▼ Convert to SPL2 Save As ▼ Create Table View Close

host="Win-Serv-22" sourcetype= *
| stats count by host sourcetype

Time range: Last 24 hours ▼  

✓ 1,756 events (8/25/26 11:00:00.000 AM to 8/26/26 11:29:38.000 AM) Job ▼ || ↶ ↷ ⬇ 💡 Smart Mode ▼

No Event Sampling ▼

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page ▼ Format ▼ Preview: On

host ↕	sourcetype ↕	count ↕
Win-Serv-22	Perfmon:Available Memory	51
Win-Serv-22	Perfmon:CPU Load	101
Win-Serv-22	Perfmon:Network Interface	184
Win-Serv-22	WinEventLog	1420

Windows 10

For this VM I will follow the same steps as with Windows Server, which is downloading and installing Splunk Forwarder.

Ubuntu-SSH

After downloading Splunk Forwarder for Linux (64x .deb) I installed it and specified the receiver (Splunk Server) IP address and port.


```

soc_lab@ubuntu-client:~/Downloads$ sudo apt install ./splunkforwarder*.deb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'splunkforwarder' instead of './splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb'
The following packages were automatically installed and are no longer required:
  libgl1-amd64-glx libglapi-mesa
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  splunkforwarder
0 upgraded, 1 newly installed, 0 to remove and 423 not upgraded.
Need to get 0 B/88.7 MB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 /home/soc_lab/Downloads/splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb splunkforwarder amd64 10.4.2 [88.7 MB]
Selecting previously unselected package splunkforwarder.
(Reading database ... 152898 files and directories currently installed.)
Preparing to unpack .../splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb ...
verify that this system has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunkforwarder (10.4.2) ...
Setting up splunkforwarder (10.4.2) ...
find: '/opt/splunkforwarder/lib/python3.7/site-packages': No such file or directory
find: '/opt/splunkforwarder/lib/python3.9/site-packages': No such file or directory
complete
N: Download is performed unsandboxed as root as file '/home/soc_lab/Downloads/splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
soc_lab@ubuntu-client:~/Downloads$

```

```

soc_lab@ubuntu-client:~/Downloads$ sudo /opt/splunkforwarder/bin/splunk start --accept-licence
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk General Terms (v4 August 2024)

```

```

soc_lab@ubuntu-client:~/Downloads$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.50.10:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: admin
Password:
Added forwarding to: 192.168.50.10:9997.
soc_lab@ubuntu-client:~/Downloads$

```

After receiver was added successfully, I specified what logs should be forwarded to splunk:

```

soc_lab@ubuntu-client:~/Downloads$ ls -lh /var/log/auth.log /var/log/syslog
-rw-r----- 1 syslog adm 56K Aug 26 12:38 /var/log/auth.log
-rw-r----- 1 syslog adm 1.6M Aug 26 12:39 /var/log/syslog
soc_lab@ubuntu-client:~/Downloads$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -sourcetype linux_secure
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/auth.log'.
soc_lab@ubuntu-client:~/Downloads$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/syslog -sourcetype syslog
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/syslog'.
soc_lab@ubuntu-client:~/Downloads$

```

Forwarding verified below:

```

soc_lab@ubuntu-client:~/Downloads$ sudo /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Active forwards:
    192.168.50.10:9997
Configured but inactive forwards:
    None
soc_lab@ubuntu-client:~/Downloads$

```

New Search SPL Convert to SPL2 Save As Create Table View Close

index=* host="ubuntu-client"
| stats count by sourcetype

Time range: Last 24 hours 🔍 🌟

✓ 4,910 events (8/25/26 12:00:00.000 PM to 8/26/26 12:45:21.000 PM) Job || 🔗 📄 ⬇️ 💡 Smart Mode

No Event Sampling

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page ✓ Format 🔘 Preview: On

sourcetype	count
linux_secure	170
syslog	4740

My next step is to install Splunk Add-on for Unix and Linux to correctly interpret and normalize Linux logs, allowing them to be effectively analysed and used by Splunk Enterprise Security.



Splunk Add-on for Unix and Linux

Install

*** Important: Read upgrade Instructions and test add-on update before deploying to production *** There are changes to default indexes and .conf changes in version 6.0 of Splunk Add-on for Unix and Linux that can break an existing installation if upgrade instructions are not followed in detail. If an existing Splunk Add-on for Unix and Linux is being upgraded, please test in a...[More](#)

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